

Question

Analyze the following chemical reactions:

1. A small amount of methane reduces manganese(III) oxide, forming manganese(II,III) oxide.
2. Nitroethane is added to aqueous sulfuric acid, and the mixture is heated to produce a monobasic acid and a 1:1 salt with nitrogen in the -1 oxidation state.
3. Iron is added to molten sodium hydroxide, yielding two elemental substances, with the iron transforming into a solid with a spinel structure.
4. Phenylhydrazine and AtI are dissolved in a $KI - I_2$ solution to produce astatobenzene.
5. Sulfur dioxide is bubbled into a black suspension of $Cu(SCN)_2$, converting the system into a white suspension.
6. Arsenic-bearing beudantite $PbFe_3(AsO_4)(SO_4)(OH)_6$ reacts with calcium bicarbonate dissolved in mine water, transforming into four minerals: scorodite ($FeAsO_4 \cdot 2H_2O$), goethite ($FeOOH$), cerussite ($PbCO_3$), and gypsum ($CaSO_4 \cdot 2H_2O$).

Among these, which reactions have chemical equations (with coefficients in the simplest integer ratio) where the sum of the product-side coefficients is even and the sum of the reactant-side coefficients is odd? Select the option that includes **all** such reactions.

A. 2

B. 4

C. 6

D. 3,5

E. 1,4

F. 2,6

G. 1,3,4

H. 1,2,5

I. 3,4,6

J. 2,5,6

K. All of the above options are incorrect.

Answer

Correct Answer: **F**

Detailed Explanation

Reaction 1: The common oxidation product of carbon is carbon dioxide, with the equation $12\text{Mn}_2\text{O}_3 + \text{CH}_4 \rightarrow 8\text{Mn}_3\text{O}_4 + \text{CO}_2 + 2\text{H}_2\text{O}$

CHECKPOINT

1 PTS



Reaction 2: The cation of a 1:1-type salt with nitrogen at -1 oxidation state should be hydroxylamine, forming $[\text{NH}_3\text{OH}][\text{HSO}_4]$ in a strongly acidic environment. The equation is $\text{C}_2\text{H}_5\text{NO}_2 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + [\text{NH}_3\text{OH}][\text{HSO}_4]$

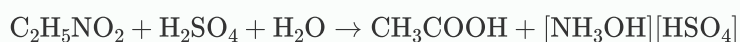
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A 1:1-type salt with nitrogen at -1 oxidation state forms $[\text{NH}_3\text{OH}][\text{HSO}_4]$ in a strongly acidic environment

CHECKPOINT

1 PTS



Reaction 3: From the spinel structure, the iron-containing species has the formula AB_2O_4 (where A,B are +2 and +3 oxidation states, respectively), indicating it is Fe_3O_4 . In a reducing environment, oxygen cannot be generated, and the

resulting elemental products are sodium and hydrogen. The equation is $3\text{Fe} + 4\text{NaOH} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{Na} + 2\text{H}_2$

CHECKPOINT

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CHECKPOINT

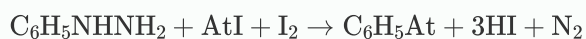
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Reaction 4: The nitrogen in phenylhydrazine converts to nitrogen gas. The equation is $\text{C}_6\text{H}_5\text{NHNH}_2 + \text{AtI} + \text{I}_2 \rightarrow \text{C}_6\text{H}_5\text{At} + 3\text{HI} + \text{N}_2$

CHECKPOINT

1 PTS



Reaction 5: The solid in the white suspension is $\text{Cu}(\text{SCN})$, and sulfur dioxide is converted to sulfuric acid. The equation is $2\text{Cu}(\text{SCN})_2 + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{CuSCN} + 2\text{HSCN} + \text{H}_2\text{SO}_4$

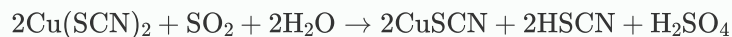
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The solid in the white suspension is $\text{Cu}(\text{SCN})$

CHECKPOINT

1 PTS



Reaction 6: Using elemental conservation, the equation is directly written as:
 $\text{PbFe}_3(\text{AsO}_4)(\text{SO}_4)(\text{OH})_6 + \text{Ca}(\text{HCO}_3)_2 + \text{H}_2\text{O} \rightarrow \text{FeAsO}_4 \cdot 2\text{H}_2\text{O} + 2\text{FeOOH} + \text{PbCO}_3 + \text{CaSO}_4 \cdot 2\text{H}_2\text{O} + \text{CO}_2$

CHECKPOINT

1 PTS



Reactions 2 and 6 have an even sum of product coefficients and an odd sum of reactant coefficients. The answer is F.

CHECKPOINT

1 PTS

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